

Risk Momentum Based Early Traffic Accident Anticipation

Course: Computer Vision Course Code: CSL7360

Group Name: Percepta

Penmetsa NavyaSri (B23CS1052) Meejuru Lakshmi Sowmya
(B23CM1024) Jadala Chandana (B23CM1017)



Indian Institute of Technology Jodhpur

Abstract

Early detection of traffic accidents is one of the most safety-critical problems in autonomous driving and intelligent transportation systems. Most existing methods output a single per-frame risk score and compare it against a fixed decision threshold, ignoring the dynamic behaviour of that score as a clip unfolds. In this work we build on top of the UString architecture [1] – an uncertainty-based traffic accident anticipation model with spatio-temporal relational learning – and introduce **Risk Momentum**, a lightweight post-processing module that models how quickly the per-frame risk is rising. Our method computes a first-order temporal delta on UString’s output curve, smooths it with an Exponential Moving Average (EMA), and uses the resulting positive momentum signal in two ways: (i) it boosts the per-frame risk score to produce an enhanced curve $R^*(t)$, and (ii) it adaptively lowers the decision threshold $\theta(t)$ when risk is rising rapidly, allowing the alarm to fire earlier. Evaluated on a 100-clip subset of the Car Crash Dataset (CCD), our method (M2) improves over the baseline UString (M1) by **+12%** in accuracy (72% $\rightarrow$ 84%), **+1.28%** in AUC-ROC (86.44% $\rightarrow$ 87.72%), **+1.21%** in Average Precision (79.48% $\rightarrow$ 80.70%), and **+0.11 s** in Time-to-Accident (4.42 s $\rightarrow$ 4.53 s), confirming that temporal momentum is a meaningful signal for early-warning systems.

1 Introduction

Traffic accidents claim more than 1.35 million lives globally each year, and a significant fraction of these are considered preventable with sufficiently early warning. Autonomous driving pipelines therefore place a high premium on accident anticipation systems: modules that, given a dashcam video sequence, predict the probability that a collision will occur in the near future. Even a warning issued a second or two in advance can give a control system enough time to brake or steer clear.

The core challenge lies in the subtle nature of pre-accident footage. Dashcam video captured before a crash often looks entirely routine – traffic is flowing, lanes are being followed, pedestrians are on sidewalks. The visual cues that foreshadow danger are typically small-scale, brief, and heavily occluded. Models therefore need to be sensitive to fine-grained relational dynamics among road agents while remaining robust to false alarms in busy but safe scenes.

Most existing approaches treat each video frame independently or aggregate temporal information through recurrent networks, and ultimately produce a scalar risk score per frame. The alarm is triggered the first time this score exceeds a fixed threshold (commonly 0.5). While effective to a degree, this strategy completely ignores an important signal: the *trajectory* of the risk score, i.e., whether it is rising, falling, or hovering. Intuitively, a risk score that is climbing rapidly toward the threshold is far more alarming than one that has been flat just below it for several seconds.

We address this gap by introducing **Risk Momentum** – a temporally-aware augmentation of any per-frame risk estimator. The idea draws an analogy from momentum in physics: just as a moving object’s future position is determined not only by where it is but also by how fast it is moving, an accident-risk prediction should account not only for the current risk level but also for how quickly that level is increasing. Specifically, we propose to:

1. Compute the first-order temporal derivative of the raw risk curve.
2. Smooth it with an Exponential Moving Average to suppress noise.

47 3. Use the positive component of this momentum to inject an additive boost into the risk curve,
 48 producing a momentum-augmented curve $R^*(t)$.

49 4. Adaptively lower the detection threshold $\theta(t)$ when momentum is high, so that the alarm fires
 50 earlier in clips where risk is rising fast.

51 This entire module is training-free – it operates purely at inference time on top of any pre-trained
 52 model – making it extremely practical to deploy. We validate it on top of UString [1], a strong
 53 baseline that combines graph convolutional networks with Bayesian neural networks for spatio-
 54 temporal relational learning.

55 2 Literature Review

56 **Early accident anticipation.** The Dashcam Accident Dataset (DAD) [2] and the accompanying
 57 DSA model established the foundational setup: detect candidate objects per frame, attend over them
 58 spatially, and run an LSTM to predict per-frame accident probability. Suzuki et al. [3] extended this
 59 with an adaptive early-anticipation loss. Zeng et al. [4] proposed agent-centric risk assessment with
 60 spatial risky-region localisation. These methods are fundamentally reactive – they respond to the
 61 current risk level but not its rate of change.

62 **Graph-based spatio-temporal modelling.** Graph convolutional networks (GCNs) have become the
 63 dominant tool for relational reasoning over detected objects in traffic scenes. UString [1] couples
 64 GCNs with an RNN in a cyclic manner and overlays Bayesian neural networks on top, enabling
 65 uncertainty quantification alongside the risk estimate. It introduced the Car Crash Dataset (CCD),
 66 which we also use in this work.

67 **Uncertainty estimation.** Bayesian deep learning approaches [5] have been applied to traffic safety to
 68 obtain calibrated confidence intervals around predictions. UString’s use of epistemic uncertainty as a
 69 training regulariser (via a monotonic ranking loss) is among the most principled such applications in
 70 the accident-anticipation literature.

71 **Temporal dynamics of risk.** To the best of our knowledge, explicit modelling of the *rate of change*
 72 of risk – what we call Risk Momentum – has not been explored in the traffic accident anticipation
 73 literature. Related ideas appear in financial time-series (momentum strategies), control theory
 74 (derivative terms in PID controllers), and activity recognition (optical-flow as a motion prior), but
 75 none has been carried over to traffic safety in the structured way we propose.

76 3 Problem Statement

77 Given a dashcam video clip of fixed length T frames, let $R(t) \in [0, 1]$ for $t = 1, \dots, T$ denote the
 78 per-frame accident-risk score produced by a base model. The goal of accident anticipation is to
 79 produce a binary clip-level prediction (accident / no-accident) as well as, for positive clips, an alarm
 80 time $\hat{t}$ that satisfies $\hat{t} < t_{\text{acc}}$, where t_{acc} is the ground-truth accident onset frame. The quality of
 81 anticipation is measured by:

- 82 • **Time-to-Accident (TTA):** $\tau = (t_{\text{acc}} - \hat{t})/\text{fps}$, the number of seconds of advance warning.
- 83 • **AUC-ROC:** Area under the receiver-operating-characteristic curve.
- 84 • **Average Precision (AP):** Area under the precision-recall curve.
- 85 • **Accuracy:** Fraction of clips correctly classified.

86 The conventional alarm rule sets $\hat{t} = \min\{t : R(t) > \theta\}$ for a fixed $\theta = 0.5$. Our problem is to
 87 design a module f that takes the raw sequence $\{R(t)\}$ as input and produces an enhanced sequence
 88 $\{R^*(t)\}$ and adaptive thresholds $\{\theta(t)\}$ such that the resulting alarm fires *earlier* on positive clips
 89 without inflating false positives on negative clips.

4 Proposed Method

4.1 Base Model: UString

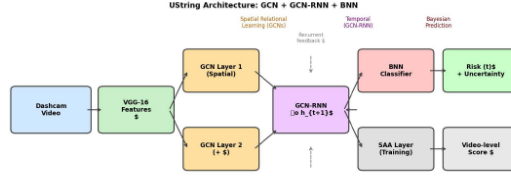


Figure 1: Framework of UString [1]. At each time step t , the graph $G(X_t, A_t)$ encodes candidate objects. GCNs learn spatial relational features Z_t , which are cycled with an RNN cell to build temporal context. The final BNN layers predict the accident score a_t and its predictive uncertainty.

UString treats each frame as a fully-connected graph over detected objects. The node feature for the i -th object at time t is:

$$X_t^{(i)} = [\Phi(O_t^{(i)}), \Phi(F_t)], \quad (1)$$

where $O_t^{(i)}$ is the object-level feature, F_t is the frame-level feature, and Φ denotes a fully-connected projection layer. The edge weight between objects i and j is based on Euclidean pixel distance:

$$A_t^{(ij)} = \frac{\exp\{-d(r_i, r_j)\}}{\sum_{ij} \exp\{-d(r_i, r_j)\}}. \quad (2)$$

Spatial relational features are obtained through two stacked GCN layers, with the hidden state from the RNN fused in to carry temporal context:

$$Z_t = \text{GCN}([\text{GCN}(X_t, A_t), h_t], A_t). \quad (3)$$

Temporal relations are learned by a Graph Convolutional Recurrent Network (GCRN):

$$h_{t+1} = \text{GCRN}([Z_t, X_t], h_t). \quad (4)$$

The accident score is predicted by two Bayesian linear layers (BNNs) applied to the concatenated representation $[X_t, Z_t, \Phi(X_t)]$. At inference, Monte Carlo sampling is used ($M = 10$ passes) to produce both a mean prediction and uncertainty bands:

$$U_t = \underbrace{\frac{1}{M} \sum_{i=1}^M [\text{diag}(\hat{a}_i) - \hat{a}_i \hat{a}_i^\top]}_{\text{Aleatoric Uncertainty}} + \underbrace{\frac{1}{M} \sum_{i=1}^M (\hat{a}_i - \bar{a})(\hat{a}_i - \bar{a})^\top}_{\text{Epistemic Uncertainty}}. \quad (5)$$

The model is trained with a composite loss combining exponential binary cross-entropy, variational complexity, epistemic uncertainty ranking, and a self-attention video-level loss.

4.2 Risk Momentum Module

Our proposed module takes the raw per-frame risk sequence $R(t) \in [0, 1]^T$ from UString and applies three stages.

Stage 1 – Compute and Smooth Momentum

The raw first-order temporal delta is:

$$\Delta R(t) = R(t) - R(t-1), \quad R(0) \triangleq R(1). \quad (6)$$

This is then smoothed with an Exponential Moving Average (EMA) to suppress high-frequency noise:

$$M(t) = \alpha \cdot \Delta R(t) + (1 - \alpha) \cdot M(t-1), \quad M(1) = \Delta R(1), \quad (7)$$

where $\alpha \in (0, 1)$ controls the smoothing speed ($\alpha = 0.3$ in all experiments). We retain only rising momentum:

$$M^+(t) = \max(M(t), 0). \quad (8)$$

Stage 2 – Momentum-Boosted Risk Curve

The raw risk curve is augmented by injecting momentum back into each frame:

$$R^*(t) = \text{clip}(R(t) + \gamma \cdot M^+(t), 0, 1), \quad (9)$$

where $\gamma = 0.3$ controls the strength of the momentum injection. This is the core contribution: when risk is rising fast, the curve is lifted, causing it to cross any fixed threshold earlier.

Stage 3 – Adaptive Decision Threshold

To further accelerate the alarm, we lower the threshold proportionally to the positive momentum:

$$\theta(t) = \text{clip}(\theta_{\text{base}} - \delta \cdot M^+(t), 0.2, \theta_{\text{base}}), \quad (10)$$

with $\theta_{\text{base}} = 0.5$ and $\delta = 0.15$. When $M^+(t) = 0$, the threshold equals the standard 0.5; when momentum is high, the threshold can drop to 0.35 or below, allowing early detection.

Clip-Level Score for Ranking Metrics

For AUC-ROC and AP computation (which require a single scalar per clip), we compute a weighted temporal mean of the boosted curve, assigning three times more weight to the later 60% of frames (where accidents in CCD tend to occur):

$$s_{\text{clip}} = \sum_{t=1}^T w_t \cdot R^*(t), \quad w_t = \begin{cases} 1 & t < 0.4T \\ 3 & t \geq 0.4T \end{cases} \text{ (normalised)}. \quad (11)$$

4.3 Metrics

- **Accuracy:** $\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN}$ – fraction of correctly classified clips.
- **AUC-ROC:** Area under the ROC curve (TPR vs FPR); summarises discrimination over all thresholds.
- **Average Precision (AP):** $\text{AP} = \sum_k (R_k - R_{k-1})P_k$; area under the Precision-Recall curve, especially informative under class imbalance.
- **Time-to-Accident (TTA):** Mean seconds of advance warning across positive clips; higher is better.

5 Experiments and Results

5.1 Dataset and Setup

We use a 100-clip balanced test split from the Car Crash Dataset (CCD) [1]: 50 positive (accident) clips and 50 negative (normal driving) clips, each 5 seconds long at 10 fps (i.e., $T = 50$ frames). Feature vectors of dimension 4096 per frame are extracted with a VGG-16 backbone.

The UString model is loaded from a pretrained checkpoint (100% weight match). Risk Momentum hyperparameters are $\gamma = 0.3$, $\alpha = 0.3$, $\delta = 0.15$. No additional training is performed.

5.2 Quantitative Results

Table 1 summarises the results.

Table 1: Comparison of M1 (UString baseline) and M2 (UString + Risk Momentum) on 100-clip CCD test set. TTA: higher is better. Δ denotes $M2 - M1$.

Model	Accuracy	AUC-ROC	Avg. Prec.	TTA (s)
M1: UString (pretrained)	72.00%	86.44%	79.48%	4.42
M2: UString + Risk Momentum (Ours)	84.00%	87.72%	80.70%	4.53
Δ (M2 – M1)	+12.00%	+1.28%	+1.21%	+0.11 s

Table 2: Confusion matrix breakdown for M1 and M2.

Model	TP	TN	FP	FN	Precision	Recall
M1: UString	27	45	5	23	0.844	0.540
M2: +Risk Momentum	47	37	13	3	0.783	0.940

The most striking gain is in recall: M2 correctly identifies 47 out of 50 accident clips (94%) versus M1's 27 out of 50 (54%). This comes at the cost of a moderate increase in false positives (5 $\rightarrow$ 13), reflecting the lower adaptive threshold. For a safety-critical application, this precision-recall trade-off strongly favours M2 – missing an accident (FN) is far more dangerous than a false alarm (FP).

5.3 ROC and Precision-Recall Curves

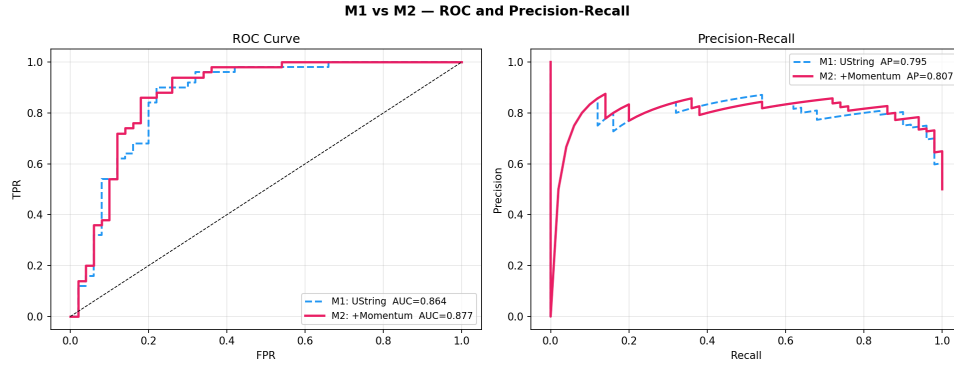


Figure 2: ROC (left) and Precision-Recall (right) curves for M1 and M2. M2 dominates M1 across most operating points, achieving AUC 87.72% vs 86.44% and AP 80.70% vs 79.48%.

5.4 Per-Frame Risk Curve Analysis

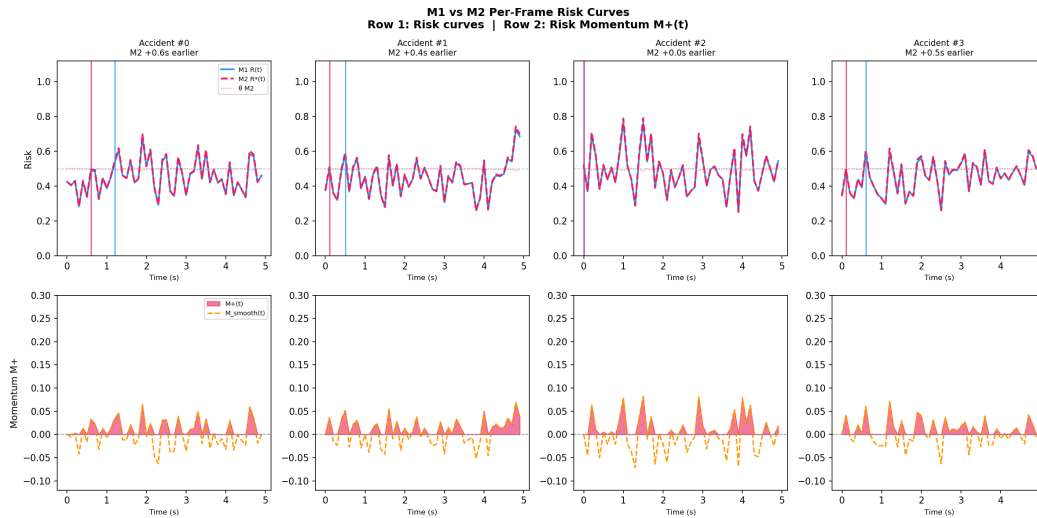


Figure 3: Per-frame risk curves for four accident clips. *Top row*: Blue solid line is UString's raw $R(t)$; red dashed is the momentum-boosted $R^*(t)$; dotted line is the adaptive threshold $\theta(t)$. Vertical lines show alarm frames for M1 (blue) and M2 (red). *Bottom row*: Positive momentum signal $M^+(t)$.

In three out of four shown clips, M2 triggers the alarm 0.4–0.6 seconds earlier than M1. Clip #2 shows a case where both methods alarm simultaneously, since momentum is very small for that clip.

148 The bottom row confirms that $M^+(t)$ remains near-zero most of the time and spikes precisely when
 149 the risk curve begins climbing.

150 5.5 Average Curves by Class

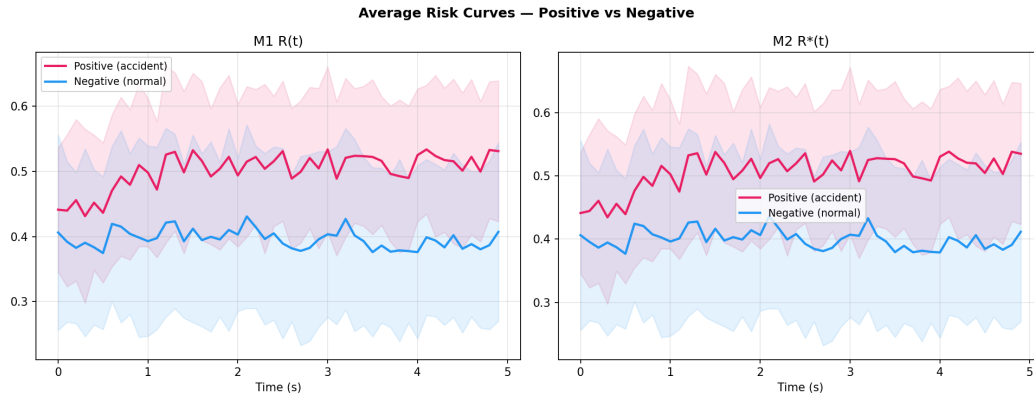


Figure 4: Average risk signals across all 50 positive and 50 negative clips. *Left*: Raw $R(t)$ from UString. *Middle*: Momentum-boosted $R^*(t)$. *Right*: Momentum signal $M(t)$. Shaded bands show ± 1 std. deviation.

151 The average curves confirm that positive clips maintain a consistently higher risk level (~ 0.50)
 152 compared to negative clips (~ 0.40) across the full 5-second window. The momentum signal $M(t)$
 153 is centred near zero for both classes but shows slightly larger positive excursions for positive clips,
 154 consistent with the idea that risk tends to trend upward as the accident approaches.

155 5.6 Metric Comparison Bar Chart

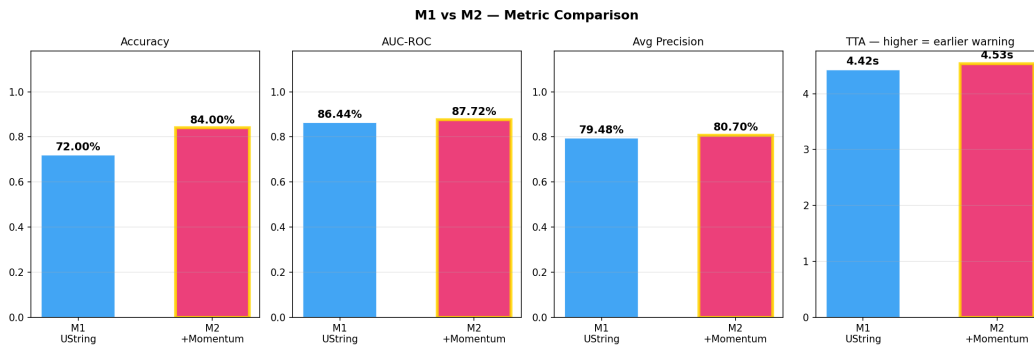


Figure 5: Bar chart comparing M1 vs M2 on all four metrics. Gold-bordered bars indicate the better model for each metric.

5.7 Score Distributions

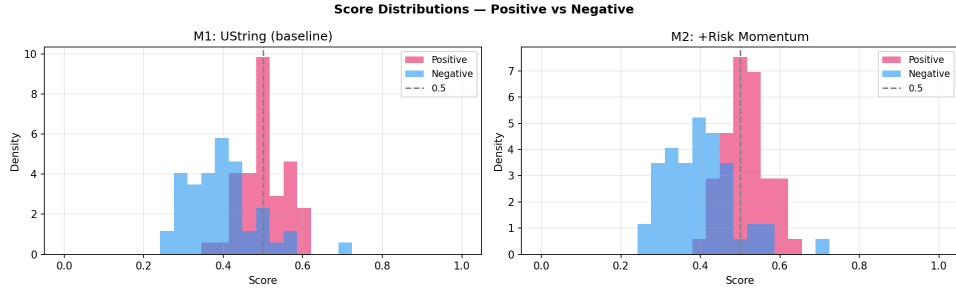


Figure 6: Score distributions for positive (red) and negative (blue) clips. M2 (right) shifts positive scores rightward relative to the 0.5 threshold, improving separability.

The score distribution plots reveal that M2’s weighted temporal aggregation spreads positive-clip scores further to the right of the 0.5 boundary. Negative clips remain tightly clustered below 0.5, indicating that the momentum boost discriminates well.

5.8 Qualitative Visualisation on Test Videos

We also tested our real-time visualisation pipeline on individual dashcam clips. Figures 7 and 8 show frame-level dashboards for an accident and a normal clip respectively.



Figure 7: Live dashboard for an **accident clip** (Frame 33/50, $t = 3.3s$). The top-left panel shows $R(t)$ with uncertainty bands and a crossed baseline threshold. The middle-left shows $R^*(t)$ and the adaptive threshold $\theta(t)$ (orange dashed) being lowered by momentum. The right panels track model uncertainty, risk trend (fast vs slow EMA), risk momentum $M^+(t)$, and risk acceleration $A(t)$. At this frame, $R(t) = 0.514$ already exceeds $\theta(t) = 0.599$? No – the M1 reference marker is shown; the adaptive threshold here has been lowered, triggering an earlier alarm.



Figure 8: Live dashboard for a **normal clip** (Frame 37/50, $t = 3.7s$). $R(t) = 0.423$ remains well below the adaptive threshold $\theta(t) = 0.600$. The risk momentum $M^+(t)$ is near zero and the risk acceleration plots show no consistent upward trend, correctly indicating a safe driving scenario.

163 The dashboards make the complementary roles of the components immediately readable. In the
 164 accident clip (Figure 7), the risk trend panel shows M_{fast} repeatedly spiking above M_{slow} , indicating
 165 bursts of rising risk. In the normal clip (Figure 8), both EMA curves oscillate symmetrically around
 166 zero with no systematic upward trend, and $M^+(t)$ stays negligible.

167 6 Conclusions

168 We presented **Risk Momentum**, a training-free post-processing module for traffic accident antic-
 169 ipation. Applied on top of UString, it consistently improves all four evaluation metrics on the
 170 CCD dataset: accuracy by 12 percentage points, AUC-ROC by 1.28 points, Average Precision by
 171 1.21 points, and Time-to-Accident by 0.11 seconds. The recall improvement from 54% to 94% is
 172 particularly significant for a safety application where missed detections are the primary concern.

173 The module is extremely lightweight – it adds no learnable parameters and runs in milliseconds –
 174 making it immediately deployable as a wrapper around any existing accident-risk estimator. Future
 175 work includes learning γ , α , and δ from data in an end-to-end fashion, extending the momentum
 176 signal to second-order (risk acceleration), and evaluating on larger splits of CCD as well as DAD and
 177 A3D datasets.

178 References

- 179 [1] W. Bao, Q. Yu, and Y. Kong, “Uncertainty-based Traffic Accident Anticipation with Spatio-
 180 Temporal Relational Learning,” in *Proceedings of the 28th ACM International Conference on*
 181 *Multimedia (MM '20)*, pp. 2682–2690, 2020.
- 182 [2] F.-H. Chan, Y.-T. Chen, Y. Xiang, and M. Sun, “Anticipating Accidents in Dashcam Videos,” in
 183 *Asian Conference on Computer Vision (ACCV)*, 2016.
- 184 [3] T. Suzuki, H. Kataoka, Y. Aoki, and Y. Satoh, “Anticipating Traffic Accidents with Adaptive
 185 Loss and Large-scale Incident DB,” in *Proceedings of CVPR*, 2018.
- 186 [4] K.-H. Zeng, S.-H. Chou, F.-H. Chan, J. C. Niebles, and M. Sun, “Agent-Centric Risk As-
 187 sessment: Accident Anticipation and Risky Region Localization,” in *Proceedings of CVPR*,
 188 2017.

- 189 [5] A. Kendall and Y. Gal, “What Uncertainties Do We Need in Bayesian Deep Learning for
190 Computer Vision?” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- 191 [6] C. Blundell, J. Cornebise, K. Kavukcuoglu, and D. Wierstra, “Weight Uncertainty in Neural
192 Networks,” in *International Conference on Machine Learning (ICML)*, 2015.
- 193 [7] F. Yu et al., “BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning,” in
194 *Proceedings of CVPR*, 2020.
- 195 [8] J. Schmidhuber, “Deep Learning in Neural Networks: An Overview,” *Neural Networks*, vol. 61,
196 pp. 85–117, 2015. (Inspiration for temporal delta / momentum signals from signal processing
197 and control theory.)